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ORIGINAL ARTICLE

Renal cell cancer after kidney transplantation

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Abstract

Purpose This study aims to identify modifiable risk factors for de novo renal cell carcinoma (RCC) after kidney transplantation in a matched-pair approach matching for unmodifiable factors.

Patients and methods One thousand six hundred fifty-five adults who underwent kidney transplantation in the period 1 January 2000 to 31 December 2012 were analyzed. Patients with RCC after kidney transplantation were matched in a 1:2 ratio with those without RCC using the indication for transplantation, age at transplantation (± 10 years), recipient sex (male/female), number of received transplants, living organ donor transplantation (yes/no), and time of follow-up in days as matching criteria. The paired *t* test was used to compare continuous variables and the Cochran-Mantel-Haenszel test for categorical variables. Multivariable conditional logistic regression modeling was used to identify independent risk factors for RCC.

Results In matched-pair analysis, a total number of 26 incident cases with RCC after kidney transplantation could be matched. Post-transplant RCC was significantly associated with longer durations of pre-transplant hemodialysis ($p = 0.007$) and post-transplant immunosuppression with cyclosporine ($p = 0.029$) and/or mycophenolate mofetil ($p = 0.020$) and with larger proportions of post-transplant time on mycophenolate mofetil ($p = 0.046$) and/or prednisolone medication ($p = 0.042$). Multivariable conditional logistic regression modeling revealed a significant risk increasing multiplicative factor interaction between the duration of pre-transplant dialysis (years) and the time of prednisolone usage (percent/100). Cyclosporine A usage and mycophenolate mofetil usage were also revealed as independent, significant risk factors for RCC development.

Conclusions Longer pre-transplant dialysis, cyclosporine-based protocols and/or intensified immunosuppression with additional mycophenolate mofetil, and larger proportions of time of prednisolone treatment during follow-up increase de novo RCC risk.

Keywords Renal cell cancer · Renal transplantation · Immunosuppression · Hemodialysis · Cyclosporine

Introduction

Kidney cancer is the 7th and 12th most frequent cancer type in men and women in Germany, respectively. In Germany, there are about 15,000 new cases of kidney cancer per year of which 90% are renal cell carcinoma with a 2:1 male:female ratio. Approximately three quarters of the cases are diagnosed in T1 or T2 tumor stages. Observed 5-year survival rates of all cancers of the kidney are reported to be between 65% for male and 69% for female patients. The incidence rate is estimated to be 16.4 and 7.9 per 100,000 people for men and women, respectively [1, 2].

Approximately 90% of patients with kidney cancer have renal cell carcinoma (RCC) of which 75% are clear cell carcinoma, approximately 10% papillary carcinoma (type 1 or 2), and 5% chromophobe carcinoma [1, 3, 4]. The remaining 10% of RCC are a hodge-podge of rare tumor types originating from nephrons or cells of the

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collecting system, for example, carcomatoid variants of RCC, carcinoma of the collecting ducts of Bellini, renal medullary carcinoma, or mucinous tubular and spindle cell carcinoma [3, 5].

Reported risk factors for the development of RCC include higher age, male gender, positive family history [6], smoking [6–8], arterial hypertension [6–10], obesity [6–11], end-stage renal disease [12], chronic kidney disease [13], renal cystic disease [13], time on dialysis [12], urolithiasis [10], and viral hepatitis [7]. Diabetes mellitus and the metabolic syndrome have also been linked to increased RCC risk [7, 9, 10, 14].

Several studies demonstrated that patients after solid organ transplantation also have an increased risk for the development of RCC [1, 15–22]. A transplant registry study from the UK reported 155 kidney cancer cases after transplantation but only 19.5 cases would have been expected based on the data of the general population, thereby indicating a sex and age group standardized incidence ratio (SIR) of 7.9 [23]. Epidemiological data from the USA showed that kidney transplant recipients aged > 35 years have the highest risk for kidney cancer in all solid organ transplant recipients [24]. A registry study reported a 4.65-fold elevated risk for kidney cancer in transplant recipients, with the highest SIR for kidney recipients (SIR 6.66) [25]. A higher proportion of papillary RCC was found in solid organ transplant recipients when compared to the general population [26, 27]. RCC after transplantation is typically diagnosed in younger individuals as compared to the general population [28].

Increased risk of RCC after transplantation, not necessarily kidney transplantation, has been attributed to long-term immunosuppression. However, the dimension of post-transplant RCC risk also depends on the type of transplanted organ and different levels of intensity of immunosuppression. The cancer risk is the highest after kidney transplantation which may also be due to factors associated with different known risk factors, such as end-stage renal disease and/or hemodialysis [23, 29, 30], and is mostly detected in an early stage [31].

The aim of this study was to identify modifiable risk factors for the development of de novo RCC in patients after the first year of kidney transplantation. Fixed risk factors are those which cannot be changed or modified, such as age, sex, the indication for transplantation, the duration of follow-up, and the number of performed transplants. Therefore, these unchangeable risk factors were used as matching criteria in matched-pair analysis. This approach purposefully eliminates the influence of these unchangeable risk factors from analysis and may unmask those risk factors that could potentially be influenced by patients or their doctors.

Patients and methods

Inclusion and exclusion criteria

This is a retrospective 1:2 matched-pair analysis of patients with and without RCC based on adult patients who underwent kidney transplantation in a German university hospital within the Eurotransplant community. In total, 1655 adults who underwent kidney transplantation in the period 1 January 2000 to 31 December 2012 and visited the transplantation outpatient department for regular transplant control until 31 December 2014 were included. Patients with diagnosis of more than one separate RCC, e.g., in the left or right native kidney and/or the transplanted kidney at different time points, were counted as separate incident cases of de novo RCC for the purpose of subsequent analysis. Patients with another transplanted organ besides the kidney were excluded. Another five incident RCC diagnoses in five patients within the first year after kidney transplantation were excluded to eliminate cases with potentially undetected concomitant RCC at the time of transplantation, because this study focuses on modifiable risk factors for de novo RCC. As Denton et al. (2002) have shown, there is a rather high rate of occult renal cell carcinoma in patients who underwent kidney transplantation at the time of transplantation [32]. Keeping this in mind, it is necessary to choose a cutoff after the date of transplantation for the definition of de novo RCC. In our center, every patient who underwent kidney transplantation undergoes cancer surveillance with once yearly ultrasound examinations of the native kidneys after transplantation. To exclude occult pre-existing RCC, de novo RCCs were defined in this study as RCCs detected after the first year when every patient has had at least one ultrasound examination after transplantation excluding occult RCC.

Patients flow through the study is shown in Fig. 1.

Matched-pair analysis

Twenty-six separate RCC incident cases were each matched with two cases without RCC during the same length of follow-up applying the following matching criteria: age at transplantation, sex, the indication for transplantation, the number of transplantations until development of de novo RCC and living donor transplantation or deceased kidney transplantation. The indication for transplantation in this study is defined as the disease which is the cause for terminal renal failure finally leading to kidney transplantation. In total, we were able to identify 52 matched pairs. For the purpose of matching, a maximum in the difference of age between matching partners at transplantation was predefined as 10 years. The minimal duration of post-transplantation follow-up for non-RCC cases

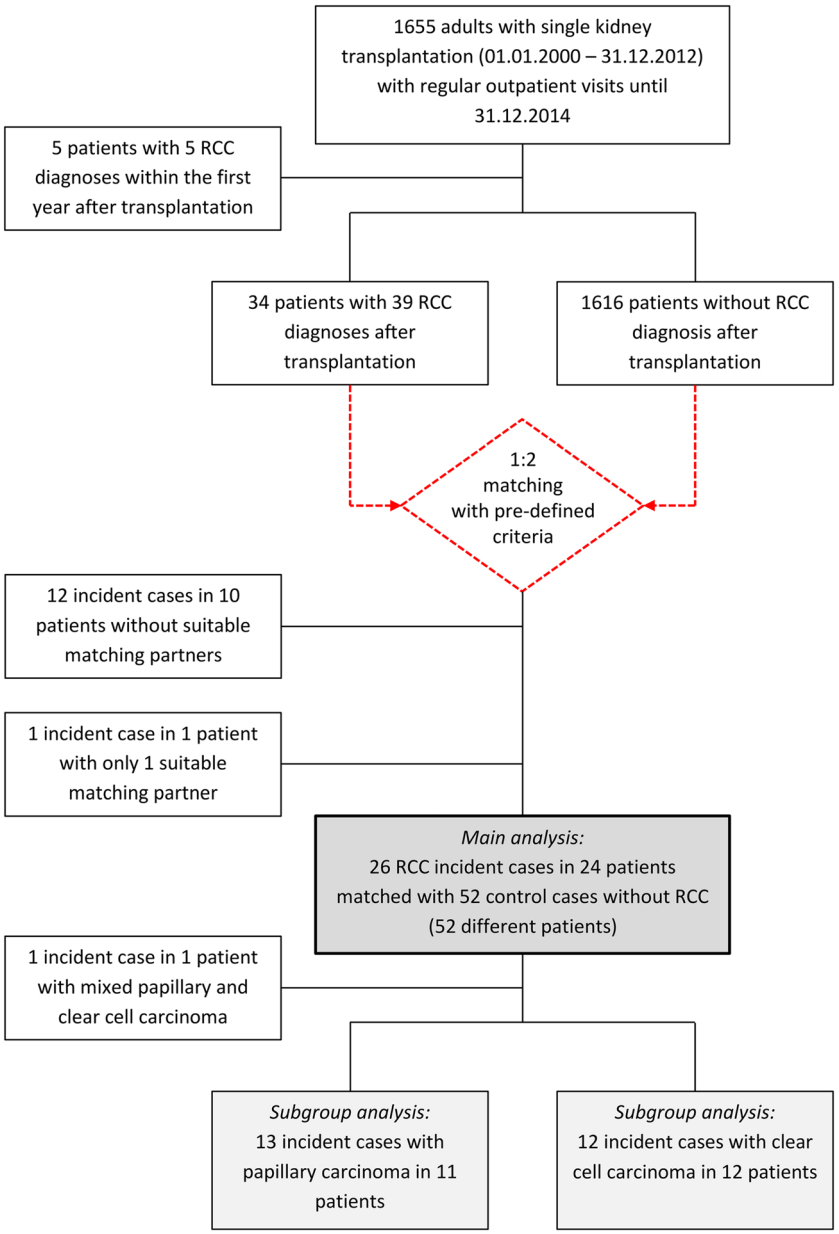


Fig. 1 Patients flow through the study

was matched to the time from transplantation until cancer development in the matching RCC incident cases. End of follow-up for RCC incident cases was defined by the last contact to our outpatient clinic closest to the actual date of RCC diagnosis. For their matching non-RCC partners, end of follow-up was defined by the date of contact to our outpatient clinic that enabled the closest match of follow-up time to the corresponding RCC incident case.

Definition of investigated variables

Quantitative data on immunosuppressive therapy of our patients was collected from hospital discharge after transplantation until the end of predefined follow-up. The duration of given immunosuppressive drugs and all changes of the type of immunosuppression were analyzed and percentages of the follow-up time with either cyclosporine-based or tacrolimus-based immunosuppression were calculated. The glomerular filtration rate (GFR) was calculated using the Cockroft-Gault formula.

Subgroup analysis after matching

The study population was also submitted to subgroup analysis for cases with clear cell carcinoma and those with papillary carcinoma as the most frequent pathological categories of RCC.

Statistical methods

Twenty-six separate RCC incident cases were each matched with two cases without RCC leading to a total of 52 matched pairs. Continuous variables between incident and control cases were compared after matching using the paired *t* test while the *p* values for differences in binominal variables were calculated using the Cochran-Mantel-Haenszel test for matched data in retrospective studies. The matched cohort was used to perform multivariable conditional logistic regression analysis using those variables that demonstrated a difference between incident and control cases in descriptive statistics with p values < 0.1. The stepwise backwards likelihood elimination method was used to detect significant independent risk factors for the development of RCC after kidney transplantation and potential factor interactions identified by odds ratio changes > 20% during this stepwise procedure. The area under the receiver operating characteristic curve (AUROC) was assessed for the prediction of RCC with the final multivariable conditional logistic regression model. Analyses were performed using JMP® Pro software (version 11.2.0, SAS Institute Inc., Cary, NC, USA). All p values < 0.05 were considered significant.

Results

Description of RCC incident cases prior to matching

Amongst 1655 adults who underwent kidney transplantation in the 13-year period, 39 RCCs were diagnosed in 34 patients (2.1%) during follow-up after transplantation as de novo RCC. The details of the RCC incident cases are summarized in Supplementary Table 1. Nineteen of these patients with RCC were male (55.9%) and 15 patients were female (44.1%). Mean age at kidney transplantation of these patients with incident RCCs was 50.1 years (range 22–67 years). The mean time between transplantation and tumor diagnosis was 1609 days (median 1167 days, range 421–3857 days).

Thirty-five RCCs were found in the native kidneys of the recipient, 15 (38.5%) in the left and 20 (51.3%) in the right native kidney, while four (10.3%) RCCs were detected in the transplanted kidney. Tumor types were diagnosed as clear cell carcinoma in 19 incident cases (48.7%) and papillary carcinoma in 16 incident cases (41.0%). In one of the latter patients, four tumors were detected in one kidney with three tumors classified as papillary carcinoma and one tumor classified as oncocytoma. The remaining four RCC incident cases (10.3%) were mixed tumors containing small cell or cystic carcinoma. According to the UICC classification, 33 incident RCCs were staged as stage I (84.6%), one incident RCC as stage II (2.6%), and five incident RCCs as stage III (12.8%).

1:2 matching results

In matched-pair analysis, a total number of 26 incident cases with RCC after kidney transplantation could be matched (66.7% of all incident cases); 15 of these matched incident RCCs occurred in males (57.7%) and 11 in females (42.3%). Ten patients with a total of 12 RCC incidences had to be excluded from analysis due to a lack of suitable matching partners. One of these excluded RCC incident cases occurred after living donor transplantation while one other incident case after living donation could be included into matched-pair analysis. Another patient with one RCC had to be excluded from analysis because only one appropriate matching partner without RCC could be identified.

After matching, the mean difference of age at transplantation was − 0.12 years (standard error 0.38 years) between RCC incident cases and non-RCC cases with a mean age at transplantation of 52.6 years for the incident cases and 52.4 years for their matching partners. All additional nominal matching variables were 100% identical between RCC incident cases and controls.

In Supplementary Table 2, an overview of characteristics of the index cases in comparison to their matching partners and the whole study cohort is shown.

Matched-pair analysis for determination of risk factors for RCC development

Duration of pre-transplant dialysis

A significantly longer duration of pre-transplant dialysis was detected in matched RCC incident cases compared to their matching partners (2419 vs 2048 days, $p = 0.007$, paired *t* test), with a mean prolonged duration on dialysis of 371 days (Supplementary Table 3, Fig. 2a). The median time of dialysis in the matched incident cases was 2400 days (range 354–3395 days) compared to a median time of 2297 days (range 250–4217 days) for the matching partners.

Table 1 shows the final multivariable conditional logistic regression model which demonstrated an AUROC for the prediction of RCC > 0.700 (0.748). An independently significant multiplicative factor interaction between the duration of pre-transplant dialysis (years) and the time of prednisolone usage (percent/100) was detected (Table 1).

In the subgroup of matched clear cell carcinoma incident cases, the mean duration of pre-transplant dialysis was 565 days longer for the RCC incident cases as compared to their matched controls ($p = 0.015$, paired *t* test) (Supplementary Table 4). In contrast, the duration of pre-transplant dialysis was not significantly different between incident RCCs and matched controls in the subgroup of matched papillary RCCs ($p = 0.250$).

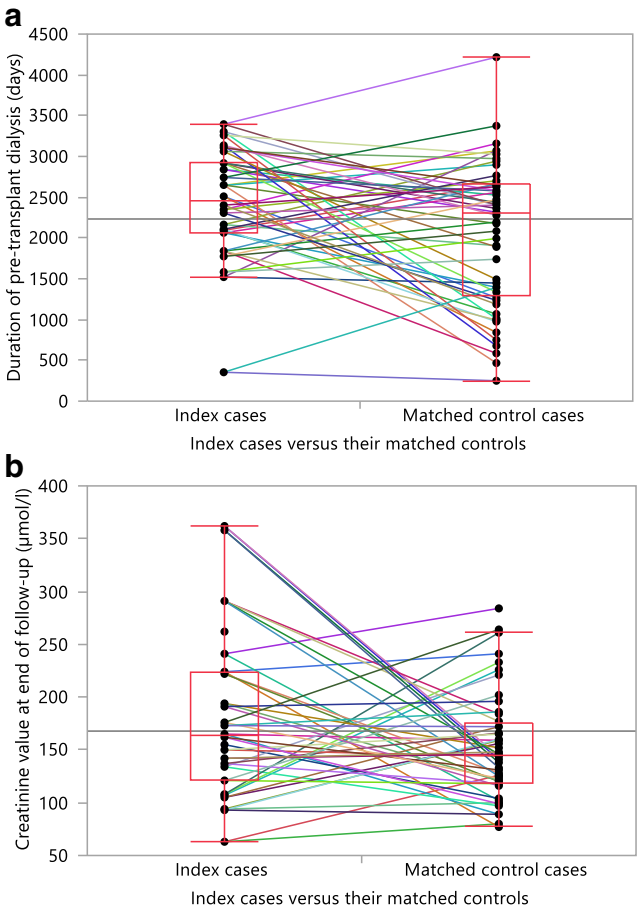


Fig. 2 Results of matched-pair analysis on the duration of pre-transplant dialysis in days **(a)** and the creatinine value at the end of follow-up **(b)** for the index cases compared to their matched controls. The duration of pre-transplant dialysis in days was significantly longer for the index cases ($p = 0.007$, paired t test) while the serum creatinine value at the end of follow-up was almost significantly different between the index cases and their matched controls ($p = 0.051$, paired t test)

Immunosuppression

Cyclosporine A (CsA) and mycophenolate mofetil (MMF) were more frequently used in the matched RCC incident cases compared to their matching partners (p values 0.029 and 0.020, respectively, Cochran-Mantel-Haenszel test; Supplementary Table 5). Comparing the proportion of time of prednisolone and MMF usage until RCC diagnosis showed that these immunosuppressants were used for significantly longer proportions of treatment time in matched RCC incident cases as compared to their matching partners (p values 0.042 and 0.046, respectively, paired t test; Supplementary Table 3). Cyclosporine A usage and mycophenolate mofetil usage could be identified as further independent risk factors for RCC in multivariable conditional logistic regression modeling (Table 1).

Subgroup analysis of clear cell carcinoma RCC incident cases revealed a significant more frequent application of CsA in matched RCC incident cases as compared to their matched controls (p value 0.002, Cochran-Mantel-Haenszel test; Supplementary Table 6). A similar finding could not be detected in the subgroup analysis for papillary carcinoma RCC incident cases when compared to their matching controls (Supplementary Table 6).

There was no statistically significant difference in usage of MMF in the clear cell carcinoma subgroup analysis between matched RCC incident cases and their matched controls (Supplementary Table 6). The papillary carcinoma subgroup analysis revealed a more frequent usage of MMF in matched RCC incident cases as compared to their matched controls which almost reached statistical significance (p value 0.059, Cochran-Mantel-Haenszel test). The comparison of the proportions of time of MMF usage demonstrated that MMF was used for a significantly prolonged proportion of treatment time in matched RCC incident cases as compared to their matching partners in the papillary carcinoma subgroup (p value 0.026, paired t test; Supplementary Table 4).

Table 1 Final multivariable conditional logistic regression model to identify independent risk factors for RCC in the matched-pair cohort. It is interesting to note that significant multiplicative factor interaction has been detected for the duration of pre-transplant dialysis (years) and the time of prednisolone usage (percent/100)

Variables	p value	Odds ratio	95% CI
Duration of pre-transplant dialysis (years) × time of prednisolone usage (percent/100)	0.013	1.268	1.044–1.540
Cyclosporine A usage (yes/no)	0.044	2.837	1.009–7.976
Mycophenolate mofetil usage (yes/no)	0.004	5.396	1.546–18.832

Increased body mass index

In the subgroup analysis of matched papillary RCC incident cases, a significantly higher mean body mass index (BMI) increase between transplantation and last follow-up of nearly 1 kg/m^2 for the RCC incident cases was identified when compared to their matching partners (p value 0.029, paired t test; Supplementary Table 4).

Serum creatinine and GFR at last follow-up

In the matched clear cell carcinoma subgroup, a significantly higher serum creatinine value was detected at last follow-up for the RCC incident cases as compared to their matching partners (p value 0.043, paired t test; Supplementary Table 4). This difference was nearly statistically significant for all matched RCC incident cases independent of their histological subtype when compared to their matching partners (p value 0.051, paired t test; Supplementary Table 3, Fig. 2b).

For the calculated GFR at last follow-up, no significant differences between matched RCC incident cases and their matching partners could be detected (p value 0.130, paired t test; Supplementary Table 3). This was also observed for the subgroup analyses of matched clear cell carcinoma and papillary carcinoma (p value 0.077 and p value 0.762, respectively, paired t test; Supplementary Table 4).

Discussion

This is the first matched-pair analysis of risk factors for RCC after kidney transplantation that has identified factors that can be influenced, such as longer pre-transplant duration of dialysis which can be shortened by earlier planning and timing of transplantation, e.g., by performing transplantation after living donation; the type and intensity of post-transplant immunosuppression which must avoid over immunosuppression; and BMI increases over time after transplantation which should be limited.

The deployed 1:2 matched-pair approach treated all individual RCC diagnoses in all patients as separate incident cases, because subsequent RCC diagnoses in the same individual after transplantation may be associated with changes of modifiable risk factors over time such as longer exposure to specific types of immunosuppression, changes in graft function at the time of diagnosis as a possible consequence of potentially avoidable rejection episodes, and/or changes of BMI due to metabolic syndrome and/or lack of physical activity. Further, all individual incident cases were matched with two different control cases leading to the matching of 26 RCC incident cases with 52 control cases in 52 different patients without RCC diagnosis during follow-up (Fig. 1).

The two patients with two incident cases were analyzed as a total of four individual incident cases matched with a total of eight different control cases according to the deployed the 1:2 matched-pair approach based on the assumption that these four incident cases may be related to different risk factor profiles. In the first patient in whom two subsequently diagnosed RCCs were analyzed as separate incident cases matched with four different control cases, the time between the diagnosis of the first and second RCC incident and the time between the diagnosis of the first RCC incident in the right native kidney and the second RCC incident in the left native kidney were 547 days (Supplementary Table 1), implicating a significantly different duration of exposure to immunosuppression. Further, the BMI changes were different (24.4 vs 23.7 kg/m^2) and graft function at diagnosis was different (GFR 58.5 vs 54.7 ml/min/1.73 m^3). Only pre-transplant variables were identical for both of these incident cases because they were observed in the same patient. In the second patient in whom two subsequently diagnosed RCCs were analyzed as separate incident cases matched with four different control cases, the time between the diagnosis of the first RCC incident in the right native kidney and the second RCC incident in the left native kidney was 28 days (Supplementary Table 1), implicating a slightly different duration of exposure to immunosuppression. Probably due to the small time difference between the detection of the first and the second RCCs in this patient, the BMI and graft function at diagnosis did not differ significantly. Pre-transplant variables were identical for both of these incident cases in the same patient.

As the focus of the study was to identify changeable risk factors for RCC that can be influenced by treatment decisions and/or patient behavior, we do not believe that it would actually be helpful to exclude those two incident cases that were identified as second, subsequent RCC diagnoses in the same two patients but in two separate native kidneys in each of these patients. Taken together, we believe that the results of this study were not biased inadequately in respect to the detection of changeable risk factors for RCCs due to the fact that we have chosen to analyze incident cases matched with control cases rather than matching patients with versus without RCC. Pre-transplant variables which were determined in this study as risk factors for RCC such as the duration of dialysis prior to transplantation which is in accordance with previous studies [12, 33–37] may be emphasized by the chosen analytical approach more by analyzing each incident case of RCC diagnosis in one patient separately. In this context, it is remarkable that the two patients with two subsequent RCC incident cases suffered in each incident from papillary renal carcinoma. These incidents were analyzed separately in matched-pair analysis and have developed in two different native kidneys in each patient.

Duration of pre-transplant dialysis

The results of the current study confirm previous findings for the first time in a matched-pair setting, indicating that longer duration of pre-transplant dialysis is a risk factor for de novo RCC development [12, 33–37]. It is striking that multivariable conditional logistic regression analysis identified a multiplicative factor interaction between the duration of pre-transplant dialysis (years) and the time of prednisolone usage (percent/100), meaning that the quantitative risk for RCC development associated with these two variables multiplies which highlights their relevance. Patients with many years of pre-transplant dialysis should therefore be preferably treated with prednisolone-free protocols with the goal to reduce their increased risk of RCC development after transplantation. We believe that this finding justifies more frequent ultrasound monitoring of patients after kidney transplantation with longer duration of pre-transplant dialysis in order to detect RCC as early as possible with the goal to enable curative treatment. This finding also suggests that longer periods of dialysis should be avoided by early kidney transplantation, e.g., by

early planned living donor transplantation. This observation sheds additional light on the value of preemptive transplantation before dialysis becomes a necessity. Preventive nephrectomy of native kidneys without function is a further option that could be discussed to prevent de novo RCC in patients' native kidneys in individuals with an increased risk for RCC. However, the latter approach should be balanced against the inherent surgical risks of bilateral nephrectomy. In this context, it is interesting to note that occult RCCs were found in the native kidneys of renal transplant recipients in a previous study in 4.2% of cases after performing simultaneous unilateral nephrectomy in a series of 260 kidney transplantations [32].

It appears noteworthy to mention that only 2/39 RCC incident cases (5.1%) were found after living donor transplantation while all others (94.9%) were associated with deceased donation. Both of these patients after living donor transplantation were not preemptively transplanted with durations of 222 and 354 days of pre-transplant hemodialysis at the time of transplantation. These findings, which are in line with findings from the US renal data system [34], highlight the importance of a shorter duration of dialysis since significantly more patients receive preemptive transplants when a living donor is available. This suggests that dialysis itself is a relevant risk factor for the development of RCC.

Acquired cystic kidney disease (ACKD) has been shown to be associated with a higher risk of RCC in renal transplant recipients [30]. Schwarz and co-workers have postulated that a reduction of RCC risk in transplanted patients in comparison to patients under dialysis can be achieved by reducing the incidence of ACKD [30]. A Korean study which compared RCC in transplanted patients with those under dialysis showed a significantly lower incidence of ACKD in the transplanted cohort [38]. Unfortunately, due to a lack of available data and a comparatively small sample size, we were not able to analyze the influence of ACKD on the risk of RCC after transplantation in the current study. Goh et al. found an association of RCC in renal transplant recipients with native renal cysts and the duration on dialysis [39].

By choosing age at transplantation as one of the matching criteria, we were able to exclude a possible age-based bias, because the duration of dialysis as a risk factor for the development of RCC after transplantation may be influenced by age at transplantation. The results of this study demonstrate clearly that the duration of dialysis is a risk factor for de novo RCC independent of patients' age.

Immunosuppression after kidney transplantation as a risk factor for RCC

This study evaluated risk factors for RCC independent of the time of follow-up after transplantation which equals exposure time to immunosuppression. A transplantation-specific risk factor which can be modified during treatment is the type of the deployed immunosuppressive regime and its intensity. This study demonstrated that CsA is associated with a higher risk of RCC in all types of RCC when controlling for follow-up time in matched-pair analysis. The final multivariable conditional logistic regression model demonstrated further that cyclosporine A usage and mycophenolate mofetil usage are also independent, significant risk factors for RCC development as has been suggested in matched-pair analysis (Table 1). This finding may, at first glance, be in contrast to a French study that reported a significantly higher prevalence of RCC in a transplant recipient population without any statistical association to immunosuppressive therapy [19]. This controversial finding may be caused by neglecting the follow-up time in the French study as a consequence of the deployed univariate study design without a matched-pair or multivariable analytical approach.

The observation of a CsA-mediated increase in RCC risk independent of follow-up time in this study fits to the observation of a Turkish study in which a significantly higher cancer risk was seen under therapy with CsA in comparison to tacrolimus and sirolimus [40]. Furthermore, a Japanese study demonstrated a higher risk for RCC in patients treated with CsA after renal transplantation as compared to those who received different immunosuppressive protocols without CsA [41]. However, both of these studies did not control for follow-up time after transplantation as well. A reason for these observations of a higher RCC incidence may be related to the promotion of ACKD in native kidneys which has been observed before under therapy with CsA after transplantation in a previous study [42]. In this study, matched RCC incident cases had a 11.7% larger proportion of treatment time with CsA post transplantation as compared to matched controls without reaching statistical significance ($p = 0.119$, Supplementary Table 3).

For the usage of calcineurin inhibitors (CNI) as a group of immunosuppressive substances and the usage of tacrolimus, no elevated risk for RCC could be detected in the current study. However, tacrolimus was less frequently used in patients of our study compared to CsA which could limit the conclusions of this observation. The finding has to be reevaluated once more incident cases have become available. Another limitation is the usage of CNI (CsA or tacrolimus) as a standard immunosuppression in our center; therefore, we cannot expect any significant findings on CNI-based versus CNI-free immunosuppression in the RCC incident cases and their matched controls. An independent study in another center which uses both CNI-based and CNI-free immunosuppression more frequently should reevaluate these findings.

Yokoyama et al. showed cytoproliferative effects of CsA and tacrolimus which may contribute to the observed elevated cancer risk under therapy with CsA. In contrast, MMF did not show tumor cell proliferation effects in a colon cancer mouse

model [43]. In vitro inhibition of tumor cell growth was shown for the use of MMF in different carcinoma cell lines but not specifically for RCC [44]. In the same study, this observed inhibition of tumor cell growth in vitro was not reproducible in vivo using standard immunosuppressive dosages of MMF [44]. Nevertheless, MMF is supposed to inhibit carcinogenesis in transplant recipients [45], which is supported by study results in heart transplant recipients [46]. In our study, these findings could not be confirmed. Instead, we found a significant association between the development of RCC and the use of MMF in the immunosuppressive regimen. This observation may be due to several reasons.

In this study, the deployed standard immunosuppression is typically a combination of prednisolone, CNI, and MMF. This triple combination may lead to a higher overall level of immunosuppression. It is therefore uncertain whether the seen effect of MMF is based on a carcinogenic effect of MMF itself or whether MMF is a marker for higher levels of immunosuppression which has been accused to promote oncogenesis. As postulated by Penn, oncogenesis may not be a problem of a special immunosuppressive drug but may be a risk of immunosuppression itself and the level of immunosuppression [47]. This theory is supported by the finding that a longer proportion of treatment time with prednisolone and MMF was found in matched RCC incident cases when compared to their matched controls with similar follow-up times. This indicates a longer proportion of time with a higher intensity of immunosuppression in matched RCC incident cases.

MMF and sirolimus have been proposed to reduce the incidence of post-transplant malignancies [48]. Considering the results of this study, the role of MMF for renal cell carcinoma needs to be reevaluated more thoroughly. In the CONVERT trial, a reduction of cancer risk in renal transplant recipients was seen for a sirolimus-based immunosuppression protocols when compared to a CNI-based regime though the difference for renal adenocarcinoma has not been significant [49]. A meta-analysis of five clinical trials supports these results [50].

Taken together, we believe on the basis of the results of this study as well as of other studies that a CsA-based immunosuppressive regime after renal transplantation should be avoided for the time being to reduce the risk of de novo RCC development.

Body mass index and RCC risk

Obesity is known as a risk factor for RCC [6–11]. In a Chinese non-transplant population, elevated body mass index (BMI) was reported to be associated with clear cell carcinoma [10]. In this study, a pronounced increase of BMI during follow-up after transplantation was associated with an increased risk of papillary cancer after transplantation (Supplementary Table 4). This finding is a hint that the increased weight after transplantation is not only a cardiovascular risk factor but potentially also a risk factor for de novo papillary renal cell carcinoma that could be controlled by weight control. Patients should be counseled accordingly.

Increased incidence of papillary carcinoma after renal transplantation

The observed significantly higher percentage of papillary carcinoma of all RCCs after transplantation in this study as compared with the German general population [1] is in line with previous publications on kidney transplant cohorts [26, 27]. Higher incidences of papillary carcinoma were also reported for transplant patients in comparison to dialysis patients without transplantation [29], indicating a role of immunosuppression as a risk factor for this tumor entity.

It has been shown that declining renal function is associated with an increased risk for papillary carcinoma which was explained by a possible causal relationship between renal injury and papillary carcinoma [13]. In this study, higher BMI at last follow-up was associated with higher serum creatinine values at last follow-up, however without reaching statistical significance (data not shown).

Role of kidney function after transplantation for RCC risk

Although there was no significant association between the calculated GFR at last follow-up and the development of RCC in this study, we found in the clear cell carcinoma subgroup significantly higher serum creatinine levels at last follow-up in RCC incident cases as compared to their matching partners. As an indirect marker of renal function, this may indicate a role for other, perhaps still unknown noxious agents with predominant renal elimination as potential risk factors for clear cell carcinoma which were not measured and analyzed in this study. This notion may be supported by the fact that most RCCs after transplantation were detected in native kidneys which were exposed to loss of kidney function for a prolonged time prior to transplantation (Supplementary Table 1). Yet to be determined carcinogenic substances, which cannot be eliminated by hemodialysis, may cause elevated cancer risks with longer duration of dialysis.

Survival after transplantation and RCC

A large French multicenter study found smaller tumors in the transplantation cohort and a better survival after transplantation in comparison to a dialysis cohort without transplantation [29]. The reported better survival was postulated to be associated with a regular follow-up and thus earlier diagnosis of smaller stage RCCs in patients who underwent transplantation [29]. Consequently, patients with identified risk factors for de

novo RCC may benefit from even more intense non-invasive screening for RCC after transplantation.

Limitations of this study

Limitations include the relatively small group of investigated RCC incident cases and the comparatively large percentage of excluded incident cases due to a lack of adequate matching partners. The exclusion of 13 cases for matched-pair analysis may have caused a bias in results. Especially the smaller number of analyzed cases in histological subgroups of RCCs potentially limits the generalizability of the findings further. Further limitations include the retrospective single-center nature of the study. Therefore, more studies and more subjects are needed in the future.

The risk for development of de novo RCC after kidney transplantation may also be influenced by other factors like immunosuppression prior to transplantation, which is strongly dependent on the indication for transplantation. Since the indication for transplantation was one of the matching criteria, we believe that immunosuppression prior to transplantation did not influence our results significantly. Nevertheless, more detailed knowledge of unchangeable pre-transplant risk factors for development of RCC may be helpful to identify patients who need intensified cancer screening for early detection of RCC.

On this background, we are looking forward to the planned German national transplant registry to provide a greater number of cases. We advocate the inclusion of data on de novo malignancy diagnoses into such registries during follow-up monitoring.

Conclusions

We found that longer duration of pre-transplant dialysis, type of immunosuppression, and BMI increases after transplantation are modifiable and, therefore, potentially avoidable risk factors for the development of RCC in patients after kidney transplantation. These risk factors foster the clinical recommendations to strive for early, preferably preemptive kidney transplantation in order to limit pre-transplant dialysis time, to use post-transplant immunosuppressive protocols that are not based on CsA, to avoid over immunosuppression with additional MMF and/or prednisolone, and to prevent significant weight increase after transplantation. A further clinical conclusion of this study is to monitor native and transplanted kidneys more regularly and closely for RCC development, especially in patients who have these risk factors for de novo RCC, e.g., by using kidney ultrasound in order to detect RCC and initiate treatment early.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The Ethics Committee of Hannover Medical School approved this retrospective study (approval decision number 2706-2015).

Informed consent Informed consent was obtained from all individual participants included in the study. Patients provided informed consent that their data may be used for scientific purposes at the time of hospital admission which is the general policy of our institution.

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